Critical Infrastructure Security

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Unit 6— Introduction to Cyber-Physical Systems and Control Systems

Contents

Introduction to Cyber-Physical Systems and Control Systems

Introduction to Cyber-Physical Systems and Control Systems
- Introduction
- Basic Control Concepts
- Open and Closed Loop Control Systems

Control Systems Architecture

Attacks on Cyber-Physical Systems and Hybrid Attacks

Part I

Introduction to Cyber-Physical Systems and Control Systems

Cyber-Physical and Control Systems

In this block we introduce some aspects of cyber-physical systems:

- Control Systems Fundamentals
- Interaction Between Computation and Physical Environment
- Modelling, Specification, and Analysis of Cyber-Physical Systems

Recommended Reading (1)

- K. Stoufer, J. Falco, K. Scarfone:
  NIST Special Publication 800-82 Rev. 3: Guide to Industrial Control Systems (ICS) Security
  U.S. National Institute of Standards and Technology (September 2023)

  Chapter 2
- P. Albertos, I. Mareels:
  Feedback and Control for Everyone
  Springer-Verlag (2010)

  Background reading on control systems
- J.W. Polderman, J. C. Willems:
  Introduction to Mathematical Systems Theory: A Behavioral Approach
  Springer-Verlag (1998)

  Far more detail than required, but freely available

Motivation

Most CI (apart e.g. from financial services) either consist of or have a substantial physical aspect

These physical assets are in turn usually produced or controlled using some automated systems

- Moreover, beyond critical infrastructures the majority of CPUs is now in so-called **embedded systems**, associated with physical devices

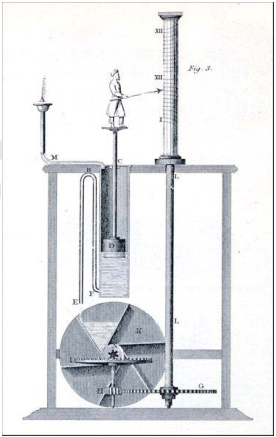
Unlike in “pure” information systems, damaging or disrupting such automated systems can cause physical damage up to loss of life and limb

A number of characteristics of **cyber-physical systems** are worthwhile to examine further as they differ substantially from standard considerations in information security

Control Systems
The Mechanical Days (1/3)

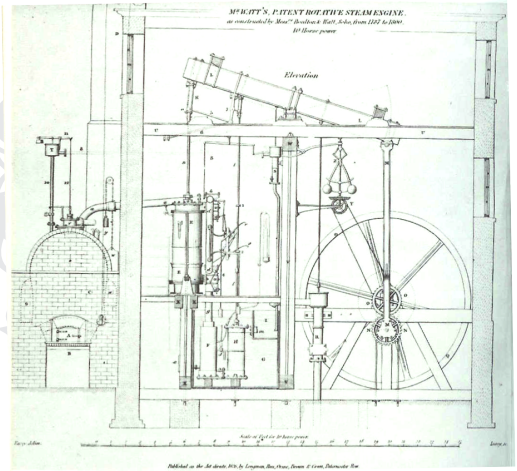
Control systems in various guises have been employed for a long time:

- A float-based mechanism for a water clock (clepsydra) is attributed to Ctesibius ($\approx$ 270 BCE)
- Earlier systems for time-keeping relied simply on water flows as early as the Egyptian and Babylonian civilisations, documented from 1600 BCE



Control Systems
The Mechanical Days (2/3)

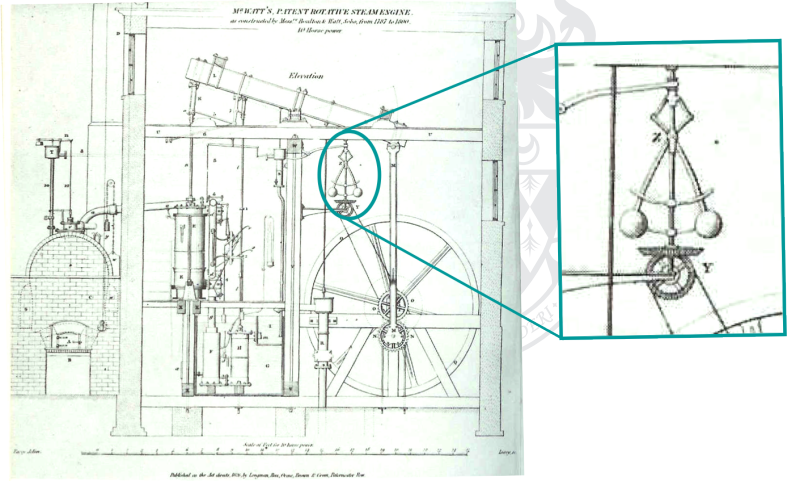
Clock-makers had built mechanisms of remarkable precision and intricacy, but it was not until the industrial revolution that these were employed elsewhere:



A steam engine design by James Watt (ca. 1780)

Control Systems
The Mechanical Days (3/3)

The flyball governor provided a simple and elegant mechanism for **feedback control**:



Control Systems Terminology (1/3)

Discussing control systems requires some specific terminology:

Process Broadly defined as an (set or sequence) operation using resources to transform inputs into outputs

Manipulated Variable The MV is a variable is the parameter (e.g. a resource) of a process that is being manipulated by the control system

Final Control Element The FCE is a component changing the value of a MV

Controller Output Signal from the controller to the FCE

Process Variable The PV is a variable (measurement) changing in response to changes in the MV

Set Point Value at which the PV is sought to be maintained

Control Systems Terminology (2/3)

Actuator Mechanism for translating a controller output (from an FCE) into physical changes, can be used synonymously with FCE

Detector A detector or sensor translates a physical parameter to a measured variable

Transducer Device receiving information in one form and converts it to another form

Transmitter Transducer responding to a MV by means of a sensing element and converts it to a standardized transmission signal that is a function of the measured variable
The sensor may or may not be part of the transmitter

Controller A device that operates automatically to regulate a controlled variable

Control Systems Terminology (3/3)

Processes exhibit time-varying behaviour, co-determined by (physical) properties of inputs, resources, and the process (i.e. its state) itself

We can now (informally) provide the following definition:

Definition (*Process Control*)

Process control is defined as the act of controlling Final Control Elements (FCEs) to change Manipulated Variables (MVs) so as to maintain Process Variables (PVs) at desired Set Points

Open-Loop Control Systems

The simplest form of control system is referred to as **open-loop** control:

- It relies only on the model of the system and
- the current state of the (modelled) system

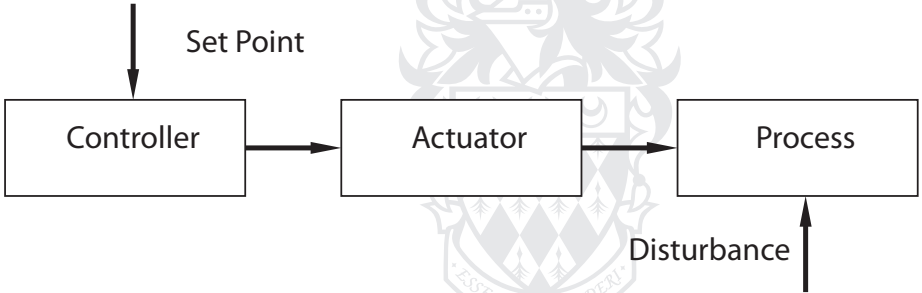
These systems are cheap to implement but do **not** rely on feedback (from sensors) to determine whether an input has resulted in the desired state change

- This makes open-loop suitable only for highly predictable and non-critical environments
- This is definitely not the case where sensors, actuators, and communication channels can be manipulated

Many processes are stable in open-loop mode, and will maintain PVs in the absence of disturbances (uncontrolled changes in process inputs or resources)

Open-Loop Control System Schema

For the basic case of a single Manipulated Variable and Process Variable, we can visualise an open-loop control system by the following schema:



Note that (e.g. external) disturbances **cannot** be taken into account by this approach

Example: Open-Loop Control System

A conventional toaster is a good example of an open-loop system:

- Calibration is essential – frozen vs. fresh vs. dry bread
- Without sensors and actuators, the colour of bread represents a number of assumptions and ignores disturbances such as voltage fluctuations

Source: Dualit

Example: Open-Loop Control System

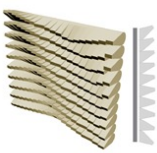
However, just when you thought it was safe. . .

Source: Electrolux

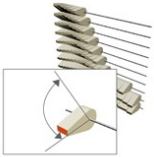
Example: Open-Loop Control System

. . . along come Inter-net-enabled toasters allowing you to download the weather forecast and print it on your toast

- Whilst this prototype was not commercialised, one hopes that similar systems will have safeguards in place that keep the “print head” from staying in contact with the toast for too long

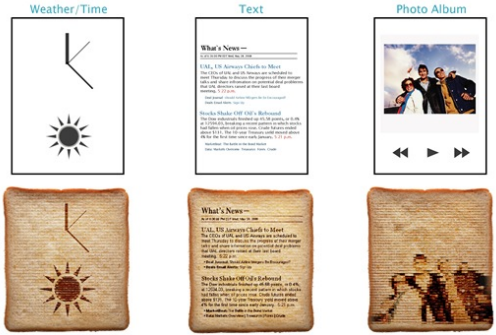


1. To display or toast the images and texts into pixels, the modules should be flexible so the heat can be trasferred by hot wires



2. Each unit module is connected to the hot wire and it's moving 30 degrees vertically so it can transfer the heat.

3. The selected images or texts into pixels that are saved on PC is toasted on the bread.



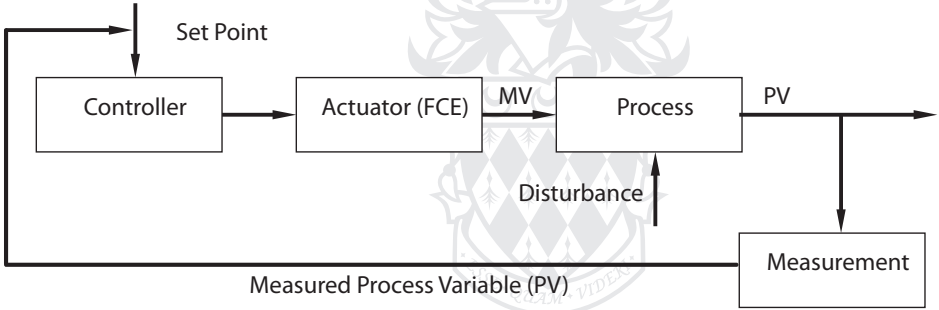
Closed-Loop Control Systems

Dealing with disturbances is usually desired or required, necessitating some type of comparison between desired behaviour from the controller and actual (observed) behaviour

- This is achieved by introducing a **feedback loop** into the control system
- A Process Variable is measured (monitored) by sensors, allowing to measure the difference (error) between desired and measured PV
- Note that there will normally exist a number of steps between the sensor recording a physical parameter and the controller

Closed-Loop Control System Schema

Showing again the case of a single **Manipulated Variable** and **Process Variable**, a simple closed-loop (feedback) control system can be visualised by the following schema:



Note that feed-forward control is also possible, but much harder: The effects of disturbances must be anticipated correctly

Control Systems Design

Assuming that

- sensors and actuators are free of faults and errors, and
- that the communication channels these are similarly free of faults and errors,

this leaves the problem of designing the controller in such a way that it will always (or under well-understood conditions) ensure that the Process Variables are in the desired range

- This is difficult enough without having to consider adversaries' interference in the control process. . .

Unlike in the examples provided, real control systems usually operate on multiple variables at the same time, which may be directly or indirectly coupled

Control Systems Design Elements

A non-trivial control system will need to be designed in a number of stages:

Modelling Physical systems must be modelled and simplified so efficient control laws can be identified, often by approximation through linearisation

Analysis Steady-state and transient time response and frequency response must be considered together with stability and robustness of the control law

Design Ultimately the control system must be feasible and cost-effective to realise

Dynamic Behaviour

Simplification is a necessary elemenmt in dealing with physical environments, but is limited by a number of considerations — a key problem is that of **process dynamics**:

- Unlike in pure IT systems, response behaviour is not instantaneous
- There is typically a dead time before a process variable starts to change, with a further lag before the final value of the process variable is achieved
- Similarly, feedback loops cause delays, requiring the consideration lead time to achieve the desired effect

Another important concept in control systems is the **gain** of a component:

- The process variable will change (usually) proportionately to the manipulated variable; this dimensionless constant k is called the gain

Modelling Physical Systems (1)

Any model can only be an approximation, but the exact choice of abstraction and simplification will vary based on the precise needs of the control law

- In the simplest case, a controlled system is well-understood and can be described using linear models with good precision
  - Mechanical systems typically fall into this category provided they are not operating outside well-understood parameters
- While non-linear systems can sometimes be described explicitly, this is frequently avoided
- Linear time-invariant (LTI) represent the most benign case, time-varying systems have parameters changing over time and must be captured explicitly
- Controlled systems are not always fully understood, or may depend on parameters (e.g. inputs) that are incomplete in their characterisation

Modelling Physical Systems (2)

As a result, models will exhibit a number of different errors

This is immediately relevant for the control system's function and efficiency, but is not limited to this:

- The ability to determine whether a process change is genuine, caused by a limitation or flaw in the model, caused by a fault in a sensor or actuator, or by a deliberate attack is similarly affected

Recall that a control system's role is essentially to detect deviations (errors) from the (current) target state:

- The mechanism for compensating for disturbances will also capture other types of errors

In addition to these considerations, there will normally be a need for explicit **state** or **mode change** modelling

- For more complex processes or where parameters change, state space models are required

Analysing Control Systems (1)

Real processes and their control systems will not always be possible to analyse exhaustively

- Instead, only well-defined parameter envelopes are often considered, and even there inputs will have some random influences and cannot be expressed analytically

Of particular interest is what happens when a deviation from a steady state (typically an a-periodic signal) occurs:

- The control system must respond to such a stimulus or internal state change
- Similar to the dynamic behaviour described for inputs, the dynamics of control outputs must also be captured explicitly

Analysing Control Systems (2)

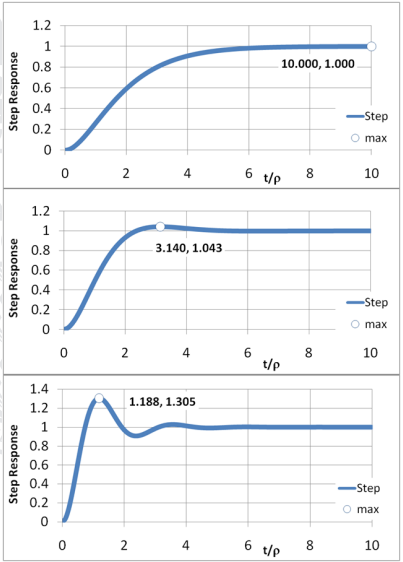
For a given state change (step function, transient, etc), the control signal in response will normally exhibit some undesirable behaviour:

- The process variable may overshoot or undershoot, requiring further actions
- Such corrections will often result in (themselves undesirable) oscillations
- Oscillations and generally the size of the control signal need to be limited; this is often referred to as the **damping** characteristics
- Stronger damping is not free, it limits the speed and magnitude with which the control system can respond

Analysing Control Systems (3)

The damping influences the **settling** period (i.e. before the process variable reaches a tolerance interval again) and the amplitude of the error before the PV settles

The resulting oscillations are also referred to as ringing



Control Limitations

Gain Constraints and Control Responses

Requirements for the magnitude of ringing and other constraints as well as the limitations of the controller itself will limit the magnitude of the gain a control system can achieve

- This results in states from which a control system cannot recover automatically within the required time/transition constraints

Such undesirable states can not only be reached when an input lies outside the system's specification, but also when the control system cannot respond correctly

- Consider the case of delayed measurements driving the process state outside the specfication before this is detected
- Similarly, bad sensors or actuators, or a mis-represented state model in the controller can result in the controller driving the system even further from the desired state

Observability

Recall that a key challenge for a control system is to retain an accurate model of the current state of the controlled process or system

This can be formalised as observability:

Definition (*Observability*)

In a state space representation, a system is said to be observable if its current state can be determined in finite time using only its output variables for any possible sequence of control and state vectors

A slightly weaker condition is detectability, here unobservable states (modes) must be stable, so non-observable states do not require stabilisation by a control system

The formulation for linear systems is due to Rudolf Kalman, this has since been generalised

Controllability

The concept of controllability is closely related to observability, it exists in different formulations — the simplest is the analog of observability defined before:

Definition (*State Controllability*)

Complete state controllability is obtained if an external input can change the internal state of a system from any initial state to any desired finite state in a finite step sequence

A different (but not equivalent!) formulation is output controllability, i.e. to move the output from any initial condition to the desired output in a finite interval

- A state-controllable system is not necessarily output controllable, i.e. may not be able to yield some outputs
- An output-controllable system may have several equivalent states (zero dynamics) resulting in the same output

Nonlinear Control

As noted, non-linear systems have a number of problematic properties:

- They do not allow the principle of **superposition** (of linear, homogeneous signals)
- They may have **multiple, isolated equilibrium points** where linear systems have only one
- They can have a **finite escape time** if unstable
- For a periodic (e.g. sinusoidal) input, outputs may contain harmonics and sub-harmonics of different amplitudes and phase differences, a linear system will only contain the sinusoid output

Summary

This section provided material on

- Control System Fundamentals
- Interactions Between Computational and Physical Environment
- Modelling, Specification, and Analysis of Cyber-Physical Systems